

Assignment 5: Model Compression on Transformer Models

Objective

To compare different model compression methods such as pruning, quantization, and knowledge distillation using a small transformer model. The goal is to analyze how each technique affects the model's size, accuracy, and inference speed, and to determine which one is most suitable for small or resource-limited devices.

Motivation

Modern deep learning models, especially transformers, are often computationally heavy and memory-intensive, making them difficult to deploy on edge devices such as mobiles and IoT systems. This assignment is motivated by the need to make these models lighter and faster while maintaining acceptable accuracy. By comparing different compression methods, we can understand the trade-offs between performance, efficiency, and deployment feasibility, guiding future optimization choices.

Experimental Methodology

Baseline Model Training:

A small transformer model was trained on the dataset to serve as the baseline. Its accuracy, model size, and inference latency were recorded as reference metrics.

Model Compression Techniques:

- Pruning: Less significant weights were removed from the trained model to reduce parameters and computation.
- Quantization: Model weights and activations were converted from 32-bit floating point to 8-bit integers to reduce memory usage and improve inference speed.
- Knowledge Distillation: A smaller “student” model was trained to replicate the outputs of the original “teacher” transformer, achieving compression through learning transfer.

Evaluation Metrics:

Each model (baseline, pruned, quantized, and distilled) was evaluated on:

- Model Size (MB) – to measure storage requirements
- Accuracy (%) – to assess performance retention
- Inference Latency (seconds/sample) – to evaluate speed on CPU

Results

(Insert comparison table and graphs here based on your experimental results.)

Key Observations

All three compression techniques reduced model size and inference time compared to the baseline transformer.

Quantization achieved the smallest model size and fastest inference but caused a minor drop in accuracy.

Pruning offered a good balance between reduced computation and maintained accuracy.

Knowledge Distillation provided a well-balanced trade-off, retaining high accuracy with significant improvements in speed and size.

On CPU, all compressed models ran faster and required less memory, making them more practical for small devices.

Conclusion

This study demonstrates that compression techniques can significantly reduce model complexity while maintaining acceptable accuracy. Among the three, knowledge distillation proved most balanced for small devices, combining efficiency and strong predictive performance. Pruning and quantization also provided substantial benefits in model size and speed. Overall, model compression enables efficient deployment of transformer models on low-power or embedded systems without major performance loss.